Three option-informed equity strategies and a SOFR-anchored portfolio

Backtests on US market data, 2022–2024

Yutong Han

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Abstract

We backtest three equity strategies built from option-market and earnings information on 15 large US stocks, using 2022–2023 to fit and 2024 as an out-of-sample year. Trades fill at official auction prices and pay a realistic fee schedule. Option-flow stock selection, which buys stocks with net option buying and sells stocks with net option selling, earned 8.6% a year over cash in 2022–2023 and 12.9% in 2024 (Sharpe ratios 0.96 and 1.71). Earnings-announcement drift, which trades in the direction of the market's reaction to an earnings release, earned 3.4% and 12.8%. A pre-earnings premium strategy earned 2.4% in 2022–2023 and 0.4% in 2024. An equal-weight portfolio of the three, with idle capital earning SOFR, returned 14.91% in 2024 against 5.40% for SOFR alone. None of these results is statistically strong: the fitting-period t-statistics are below 2, and the three strategies were chosen from a longer list partly on their 2024 results. We therefore propose paper trading the portfolio on 2025 data before committing capital.

1 Introduction

This report covers three strategies and one way to combine them. All three trade stocks, and two of them use information from the option market.

- **Option-flow stock selection** (Section 3). Traders with private information often prefer options. When option buying in a stock outweighs option selling, the stock tends to rise the next day. We buy the stocks with the most net option buying and short those with the most net selling.
- **Earnings-announcement drift** (Section 4). Prices tend to under-react to earnings news. We trade in the direction of the stock's reaction to an announcement and hold for 20 trading days.
- **Pre-earnings premium** (Section 5). Stocks have tended to earn more in the days just before scheduled earnings. We hold stocks for the five sessions before an announcement and sell before the news.

Section 6 combines the three. Each strategy uses only part of its capital at a time, so the portfolio places all unused capital in SOFR, the overnight secured funding rate published by the New York Fed. Section 7 sets out what the evidence can and cannot support.

These three strategies are what remained from a longer search. We tested 25 strategy specifications on the same data, including several option-premium strategies, and Appendix A summarizes the ones we dropped. Appendix B describes the implementation.

2 Data, trading rules and costs

Data. The sample runs from 2022-01-03 to 2024-12-31 (753 trading days). The universe is 15 large US stocks with listed options: AAPL, AMD, AMZN, AVGO, CRM, GOOGL, JPM, LLY, MSFT, NFLX, NVDA, TSLA, UNH, WMT and XOM. We use four sources:

- intraday option trades, each with the prevailing best bid and offer;
- end-of-day option chains, with implied volatility, delta and open interest;
- equity prints from the opening and closing auctions;
- daily stock data, including split and dividend adjustments, and earnings announcement dates and times.

SPY, the S&P 500 ETF, is used for hedging. SOFR fixings come from the New York Fed.

How trades fill. All three strategies trade stocks only, at two kinds of price.

- **Closing trades** fill at the official closing price, which equals the closing-auction print on every day we checked.
- **Opening trades** fill at the primary exchange's opening-auction price. If that print is missing, the order fills at the same day's close.

Auction trades do not cross a bid–ask spread, so the main costs are fees and borrowing. A strategy starts with a $1M account. Idle cash earns the 1-month Treasury bill rate inside each strategy; the portfolio in Section 6 uses SOFR instead.

Costs. Table 1 lists the costs charged on every trade.

Table 1: Transaction costs charged in every backtest.

Item	Cost
Commission and clearing	$0.0035 + $0.0002 per share
Auction exchange fee	$0.0010 per share
SEC fee on sales	$27.80 per $1M sold
FINRA trading activity fee on sales	$0.000166 per share, capped at $8.30 per trade
Borrowing cost on short positions	0.40% a year

How we judge the results. We fix each strategy's rules using 2022–2023, the *fitting period*, and evaluate them once on 2024, the *out-of-sample year*. The 2024 run used code that was committed before 2024 data entered the simulation. We report annualized return in excess of cash, the Sharpe ratio, and a Newey–West t-statistic (Newey and West, 1987). For event strategies we also report the average return per event. Three years of data is short: an annualized Sharpe ratio estimated over one year has a standard error of about 1, so differences between strategies of that size are not meaningful.

Figure 1 shows the cumulative returns of the three strategies and of their equal-weight combination.

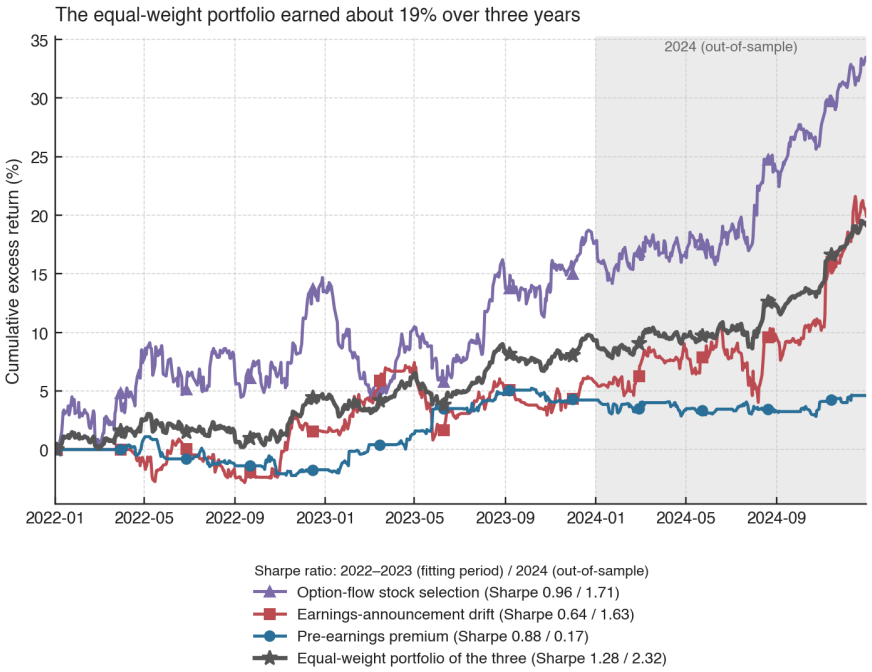


Figure 1: Cumulative return over cash for the three strategies and their equal-weight portfolio, 2022–2024. The shaded band is the 2024 out-of-sample year.

3 Option-flow stock selection

3.1 Economic intuition

A trader with private information about a company can profit more from options than from the stock, because options offer leverage and draw less attention. If such traders are active, the direction of option trading should predict the stock's next move. Easley et al. (1998) find that option volume carries information about future stock prices, Pan and Poteshman (2006) show that stocks with heavy put buying relative to call buying underperform over the following days, and Johnson and So (2012) link high option-to-stock volume to lower future returns. Implied-volatility measures built from option prices also predict stock returns (Cremers and Weinbaum, 2010; Xing et al., 2010).

3.2 Signal and trading rules

For each stock i and day t, we classify every regular-hours option trade as a buy if it printed above the midpoint of the prevailing bid and offer, and as a sell if it printed below. Each trade is weighted by the option's delta δ_k from that day's end-of-day chain, which converts it into an equivalent number of shares. The signal is the net share-equivalent buying as a fraction of all share-equivalent volume:

$$\text{flow}_{i,t} \;=\; \frac{\sum_k \delta_k \,(\text{buy}_k - \text{sell}_k)}{\sum_k |\delta_k| \,(\text{buy}_k + \text{sell}_k)}. \tag{1}$$

Because it is a ratio, the signal does not depend on how many trades the data source happens to record.

Each day we rank the tradeable stocks by flow (on average 12). Weights are the ranks minus

their average, scaled so the long side and the short side are each 50% of capital. The portfolio is therefore dollar neutral. We enter at the next morning's opening auction ($t+1$) and hold until the following opening auction ($t+2$), rebalancing every day.

3.3 Implementation

Positions trade only in opening auctions, so the costs are fees and short borrowing. Daily rebalancing turns over about 128% of capital per day, counting both sides. On a $1M account, fees average $48.8 a day and borrowing $7.9 a day. The engine is `wave3/cpp/s3_flow.cpp`.

3.4 Results

Table 2 reports the results. The strategy earned 8.6% a year over cash in 2022–2023 and 12.9% in 2024. Its returns are almost unrelated to the market: the beta to SPY is close to zero in both periods. Removing the five best days leaves 4.3% and 5.8% a year, so the result does not hinge on a handful of days.

Table 2: Option-flow stock selection, 1-day hold, after costs. Returns are annualized excess returns over cash on a $1M account.

Period	Return	Volatility	Sharpe	t-stat	Max drawdown
2022–2023 (fitting)	8.6%	9.0%	0.96	1.38	−9.1%
2024 (out-of-sample)	12.9%	7.5%	1.71	1.82	−3.1%

The signal is short-lived. Holding the same ranking for five days, with one-fifth of the book rebalanced each day, earns −2.0% in 2022–2023 and −2.9% in 2024 (Figure 2, right).

An earlier version of this strategy, with lower assumed fees, was developed on January 2022 to June 2024 and tested on the second half of 2024. It earned 7.8% a year in the fitting window (Sharpe 0.90) and 23.7% in the test window (Sharpe 3.26). A Fama–MacBeth regression (Fama and MacBeth, 1973) of the next day's return on the standardized signal gives a slope of 3.6 basis points per standard deviation in the fitting window ($t = 1.63$) and 12 basis points in the test window ($t = 2.32$), shown in Figure 2 (left).

3.5 Assessment

This is the strongest of the three strategies. It is positive in both periods, holds up after costs and without its best days, and is market neutral. Its weakness is statistical: the fitting-period t-statistic of 1.38 is well short of conventional significance, and the universe is small.

4 Earnings-announcement drift

4.1 Economic intuition

Prices tend to adjust to earnings news over several weeks rather than all at once, a pattern known as post-earnings drift (Chan et al., 1996). Measuring the surprise usually requires analyst forecasts, which our data do not include. Brandt et al. (2008) show that the stock's own abnormal

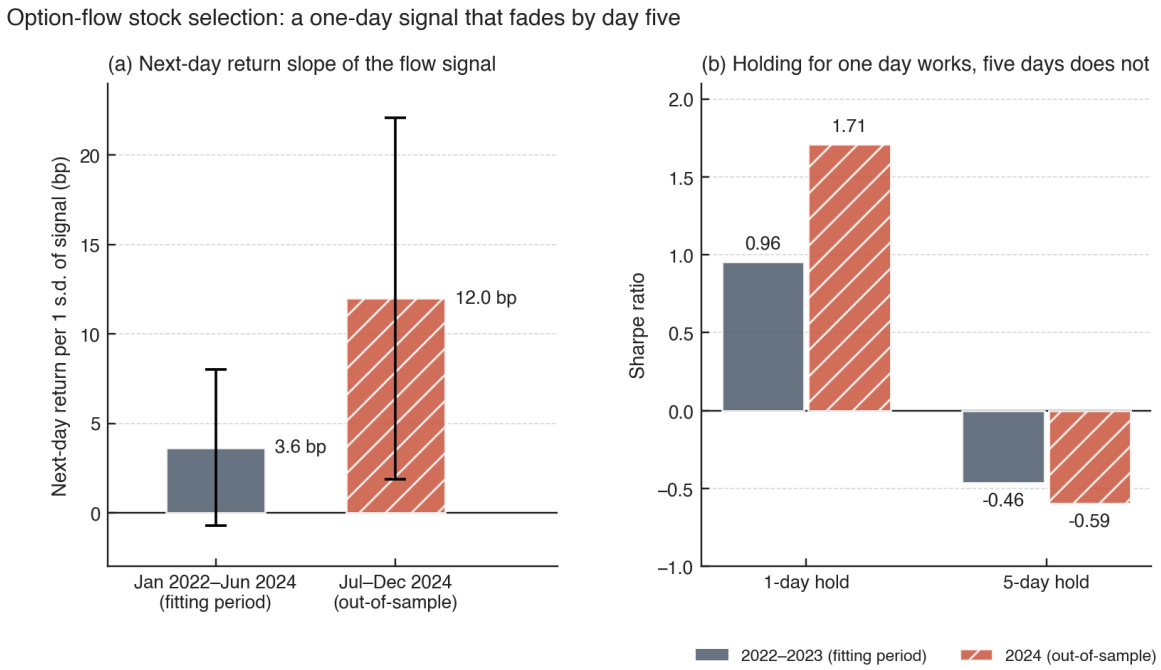


Figure 2: Left: next-day return per one-standard-deviation move in the flow signal, with 95% intervals. Right: Sharpe ratio of the 1-day and 5-day holds.

return around the announcement works as a surprise measure, and that it predicts returns over the following weeks.

4.2 Signal and trading rules

For each announcement we measure the stock's return over the reaction window, net of its market exposure:

$$\text{EAR} \;=\; r_{\text{stock}} - \beta\, r_{\text{SPY}}. \tag{2}$$

The reaction window runs from the last close before the announcement to the first close after it. For an after-close announcement, that is the announcement day's close to the next day's close. For a before-open announcement, it is the previous close to the announcement day's close. The beta β is estimated by regressing the stock's daily returns on SPY's over the previous 60 trading days, capped between 0.5 and 2 (average 1.22).

The morning after the reaction window ends, we buy the stock at the opening auction if EAR is positive and short it if EAR is negative. Each position is 10% of capital, hedged with an offsetting SPY position of β times its value. We close both after 20 trading days at the official close. Positions in different stocks can overlap.

4.3 Implementation

The data cover 163 announcements across the 15 stocks: 104 in 2022–2023 and 59 in 2024. All trades are auction or closing-price trades. The engine is `wave3/cpp/s_events.cpp`.

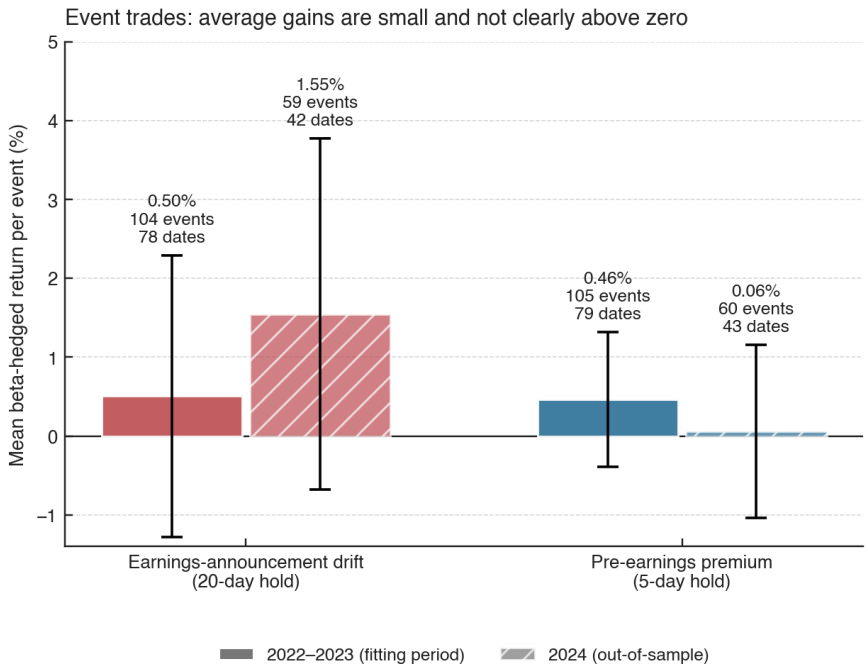


Figure 3: Average beta-hedged return per event, with 95% intervals clustered by exit date, for the two earnings strategies.

4.4 Results

Table 3 reports the results. The average hedged return per event was 0.50% in 2022–2023 and 1.55% in 2024 (Figure 3). Neither average is statistically different from zero, with t-statistics of 0.59 and 1.32. At the account level the strategy earned 3.4% a year over cash in 2022–2023 and 12.8% in 2024.

Table 3: Earnings-announcement drift, 20-day hold, beta-hedged, after costs.

	Per event			Account (annualized)		
Period	Events	Mean	t-stat	Return	Sharpe	Max drawdown
2022–2023 (fitting)	104	0.50%	0.59	3.4%	0.64	−6.2%
2024 (out-of-sample)	59	1.55%	1.32	12.8%	1.63	−6.1%

Larger reactions drift more. Restricting to announcements with $|\text{EAR}| > 2\%$ raises the average to 1.22% per event in 2022–2023 (80 events) and 1.83% in 2024 (42 events). The 2024 result is partly market exposure: although each position is hedged with a 60-day beta, the strategy's realized beta to SPY in 2024 was 0.31.

4.5 Assessment

The drift is in the direction the literature predicts and is larger in 2024. There are only about 50 to 60 events a year across 15 stocks, so the averages are noisy. The leftover market exposure in 2024 suggests the hedge needs a more stable beta estimate.

5 Pre-earnings premium

5.1 Economic intuition

Stocks have historically earned higher returns in the days around scheduled earnings announcements. Two explanations are compensation for the risk of the coming news, and buying pressure from investors whose attention the announcement draws (Frazzini and Lamont, 2007; Barber et al., 2013). The strategy holds stocks into the announcement and exits just before the news, so it takes on the pre-announcement drift but not the announcement jump itself.

5.2 Signal and trading rules

The signal is the earnings calendar. Let X be the last trading session before the announcement: the announcement day itself for an after-close release, and the day before for a before-open release. We buy the stock at the close five sessions before X and sell at the close of X. Each position is 10% of capital, hedged with an offsetting SPY position sized by the same 60-day beta as in Section 4. We assume announcement dates are known at least seven trading days in advance, as companies normally publish them.

5.3 Implementation

The data cover 165 announcements: 105 in 2022–2023 and 60 in 2024. All trades are at the official close. The engine is `wave3/cpp/s_events.cpp`.

5.4 Results

In 2022–2023 the hedged return averaged 0.46% per event ($t = 1.47$), and the strategy earned 2.4% a year over cash (Sharpe 0.88). In 2024 the average fell to 0.06% per event ($t = 0.02$), with 0.4% a year (Sharpe 0.17). Table 4 summarizes the results, and Figure 3 compares the per-event averages with the drift strategy.

Table 4: Pre-earnings premium, five-session hold, beta-hedged, after costs.

	Per event			Account (annualized)		
Period	Events	Mean	t-stat	Return	Sharpe	Max drawdown
2022–2023 (fitting)	105	0.46%	1.47	2.4%	0.88	−3.3%
2024 (out-of-sample)	60	0.06%	0.02	0.4%	0.17	−1.4%

Without the SPY hedge, the 2022–2023 average rises to 1.10% per event ($t = 2.28$). Most of that is market exposure, and the unhedged 2024 average is close to zero.

5.5 Assessment

This is the weakest of the three. The 2022–2023 result is small and mostly disappears once its five best days are removed, and the premium did not appear in 2024. We keep it in the portfolio for its low volatility and low correlation with the other two strategies. Its own evidence is weak.

6 Combining the three strategies

6.1 Setup

Each strategy uses only part of its capital at any time. Option-flow selection is fully invested but dollar neutral, and the two earnings strategies hold positions only around announcements. We therefore combine them in one account and place all capital not assigned to a strategy in SOFR. With weights w_i on strategy excess returns $x_{i,t}$ and the SOFR accrual c_t (actual/360, at the previous session's fixing), the daily portfolio return is

$$r_t \;=\; c_t + \sum_{i=1}^{3} w_i\, x_{i,t}, \qquad w_i \geq 0, \quad \sum_i w_i \leq 1. \tag{3}$$

In 2024 SOFR alone returned 5.40%; in 2022–2023 it averaged 3.37% a year.

6.2 Allocation rules

Weights are estimated on 2022–2023 only and held fixed through 2024. We compare three rules.

- **Equal weight**: one third of capital each. Simple rules like this are hard to beat out of sample, because estimated means are noisy (DeMiguel et al., 2009).
- **Inverse volatility**: weights proportional to the reciprocal of each strategy's volatility, so each contributes similar risk.
- **Markowitz**: weights that maximize the expected Sharpe ratio (Markowitz, 1952), with no strategy above 50%. Expected returns are shrunk toward a common value (Jorion, 1986), and covariances are shrunk toward a simpler structure (Ledoit and Wolf, 2004), both to limit estimation error.

Table 5 shows the resulting weights. Markowitz and inverse volatility both favor the pre-earnings premium because of its low volatility.

Table 5: Weights estimated on 2022–2023.

Strategy	Equal weight	Inverse volatility	Markowitz
Option-flow stock selection	33.3%	16.4%	13.2%
Earnings-announcement drift	33.3%	29.1%	36.8%
Pre-earnings premium	33.3%	54.4%	50.0%

6.3 Results

The three strategies have low correlations with one another, so the combination is steadier than any single strategy. Table 6 reports the results. Equal weight had a 2022–2023 Sharpe ratio of 1.28, against 0.64 to 0.96 for the individual strategies. In 2024 it returned 14.91%, or 8.71% over SOFR, with a maximum drawdown of −1.68%.

All three rules beat SOFR in 2024 (Figure 4). Equal weight did best, because it gave the most weight to the two strategies that were strongest in 2024. With one year of data, the differences between the rules are within noise. Re-estimating the Markowitz weights each month during 2024 changes little: it returns 12.51%.

Table 6: Portfolio results. Total return includes SOFR on idle capital; excess return and Sharpe ratio are relative to SOFR.

Period	Rule	Total	Excess	Volatility	Sharpe	Max drawdown
2022–2023	Equal weight	8.16%	4.54%	3.54%	1.28	−2.21%
	Inverse volatility	6.99%	3.42%	2.46%	1.39	−1.99%
	Markowitz	6.84%	3.28%	2.49%	1.32	−1.92%
2024	Equal weight	14.91%	8.71%	3.75%	2.32	−1.68%
	Inverse volatility	11.94%	6.07%	3.04%	2.00	−1.33%
	Markowitz	12.55%	6.62%	3.42%	1.94	−1.54%

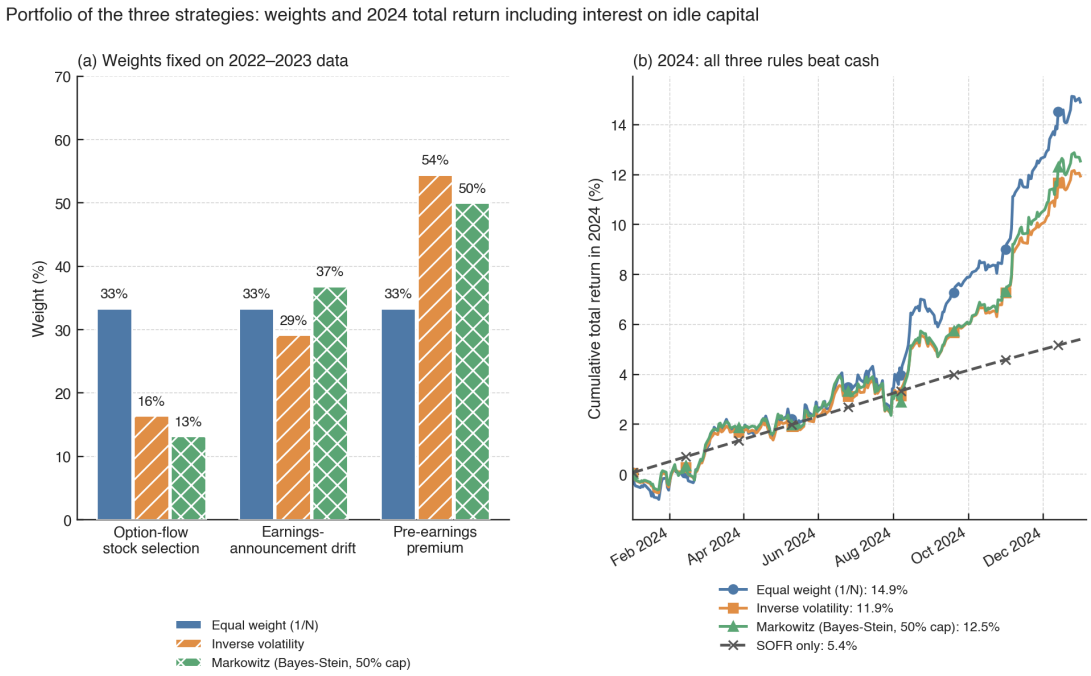


Figure 4: Left: weights estimated on 2022–2023. Right: cumulative 2024 total return of each rule, including SOFR on idle capital, against SOFR alone.

The portfolio runs at about 3–4% volatility. Scaling equal weight to a 5% volatility target needs 1.41 times leverage; in 2024 that would have returned 19.01%, with a maximum drawdown of −2.46%.

6.4 Caveat and proposal

These portfolio results overstate what an investor could have expected. We selected the three strategies after seeing their 2024 results, so 2024 is not out-of-sample for the choice of strategies. It is out-of-sample only for each strategy's rules and for the weights.

We propose equal weighting, the rule with the fewest estimated inputs, and a staged start:

1. Run the frozen portfolio on 2025 data without changes. Proceed only if it beats SOFR after costs and its drawdown stays under 2%.
2. If it passes, allocate a small risk budget, with the rest of the capital in SOFR. Stop any strategy

whose rolling 63-day Sharpe ratio falls below -1, or whose drawdown exceeds twice its 2022–2024 maximum.

This is a research proposal, not investment advice.

7 Limitations and next steps

Statistical strength. No result here is strong enough to rely on. The fitting-period t-statistics are 1.38 for option flow, 0.87 for earnings drift and 1.08 for the pre-earnings premium. The deflated Sharpe ratio (Bailey and López de Prado, 2014) adjusts for the number of variants tried, 88 across our last two rounds of testing. It is 0.13, 0.05 and 0.09 for the three strategies. A value near 1 would indicate a result unlikely to come from searching alone.

Selection. We kept these three strategies after seeing how all 25 performed, including in 2024. Each one's 2024 result is a genuine out-of-sample test of rules fixed in advance, but the choice of which three to present is not.

Data. The universe has 15 stocks, of which about 12 are tradeable on a typical day because some lack auction data. The option trade record is a sample of all trades rather than the full tape, so the flow signal is measured with noise. Fees follow a public schedule rather than executed broker statements.

Next steps. The rules for all three strategies and the portfolio are frozen in code. The natural next test is a single run on 2025 data with no changes. The option-flow signal would also benefit from a larger universe: its strength comes from ranking many stocks each day, and 15 names limits that.

A Other strategies we tested

We also tested strategies that did not reach the report's main sections, under the same costs and rules. Table 7 lists each one with its out-of-sample result and the reason we dropped it. Most failed for one of two reasons. Some paid away their edge in bid–ask spreads, because they had to buy and sell options. Others were positive in one period and negative in the other.

Table 7: Strategies tested and dropped. "Out-of-sample" is 2024 (or 2024H2 for strategies H1–H6); returns are annualized excess returns on a $1M account.

Strategy	Idea	Out-of-sample	Why dropped
Credit spreads, condors, debit verticals	Sell or buy option premium daily on SPY, QQQ, SPX	−0.4% to −4.5%	Spread costs exceeded the premium
0DTE SPXW iron fly (H1)	Sell same-day index options, hold to cash settlement	+3.0%	Negative in 2022–2023 (−3.1%)
Overnight short straddle (H2)	Collect the overnight option premium, delta-hedged	+0.9%	Spread costs four times the premium
Overnight stock drift (H3)	Buy at the close, sell at the next open	+28.8%	Result relied on a few days
Pre-earnings straddle (H5)	Buy volatility before earnings	−0.5%	Negative in both periods
End-of-day momentum (H6)	Follow the day's move into the close	−0.9%	Negative out of sample
Pre-announcement option flow (S4)	Option flow before earnings	+0.2%	No effect
28-day SPXW iron fly (S5)	Sell index volatility when the VIX curve slopes up	−8.9%	Negative in both periods
Volatility-gap straddles (S6)	Long cheap, short rich single-stock volatility	−6.9%	Negative in both periods
0DTE pin (S7)	Center the iron fly on the largest open-interest strike	−0.1%	No pin effect
Daytime reversals (S8)	Rank stocks by overnight-up, intraday-down days	+19.6%	Negative in 2022–2023
FOMC drift (S9)	Hold SPY into Fed announcements	−0.9%	Negative out of sample
Turn of the month (S10)	Hold SPY over month ends	−3.6%	Negative out of sample

As a check on the accounting, a monthly SPX put-writing strategy built with the same option code tracks the published Cboe PUT index with a monthly correlation of 0.972 and a tracking error of 2.3% a year.

B Implementation

Code. Each strategy is a C++ program that reads reduced data files, applies the rules day by day, and writes a trade list and a daily return series (`wave3/cpp/s3_flow.cpp`, `wave3/cpp/s_events.cpp`). Python scripts reduce the raw data with DuckDB and compute the statistics (`wave3/scorecard3.py`, `alloc/allocate.py`). Notebooks produce the figures.

Accounting. Stock positions are marked to the official close every day. Splits adjust share counts, and dividends are credited on the ex-date. Each strategy refuses to simulate dates after a configured end date, which is how the 2024 data were held back during development.

Checks. We reconciled the stock-position ledger by hand: price gains of $111,375, less $861 in fees, plus $797 in dividends, match its $111,311 total exactly. Rebuilding the engines from a clean copy of the repository reproduces the stored results byte for byte.

The reference details below should be checked against the published versions before citing.

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